

2×2 decomposition of TWAS discordance

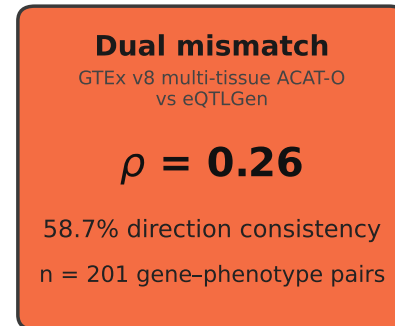
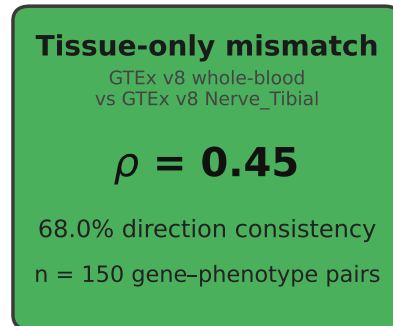
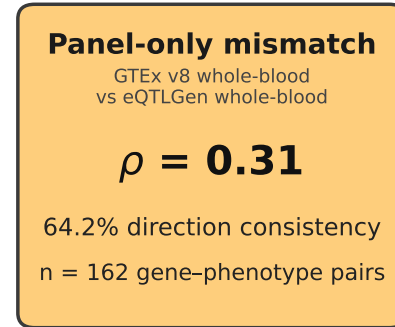
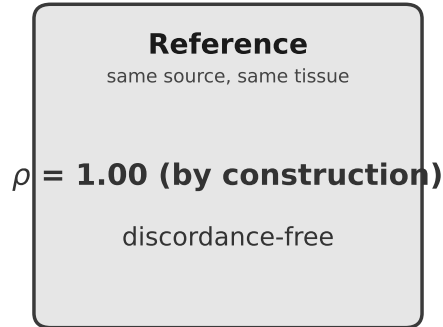
eQTL-source / panel axis

┌────────── GTEX v8 ───────────┐ ┌────────── eQTLGen ───────────┐
└────────── (same source) ───────────┘ └────────── (different source) ───────────┘

Tissue-context axis

Matched
(whole blood)

Mismatched
(Nerve_Tibial / multi-tissue)



bounds discordance
attributable to the
eQTL-source axis
(primary $\rho=0.29$, $n=102$)

Spearman ρ (higher = more concordant / less discordant)

0.20 0.30 0.40 0.50

Off-diagonal cells report Spearman ρ and direction-consistency of the corresponding pairwise TWAS comparison.
Reference (same source, same tissue) cell is discordance-free by construction. Primary GTEX v8-eQTLGen comparison ($n=102$): $\rho=0.29$.
Robustness: ordering preserved in the common-gene universe ($n=144$) and all three phenotype strata (DR/DN/DPN).